

# Git Basics Continued

## IEM 4723 Information Systems Design

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*Image transcription and some slide annotations were carried out using AI.*

- 1 GitHub Desktop: A GUI Alternative
  - Branching and Merging
  - Pull Requests
  - Forking
  - Closing the Loop: A PR from a Fork

## A GUI Alternative: GitHub Desktop

# GitHub GUI Options

## GitHub Website:

- Web-based interface for managing repositories
- Create, fork, clone repositories
- Manage issues, pull requests
- View file history and branches

## GitHub Desktop:

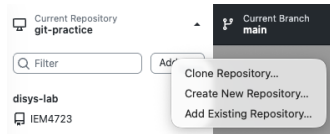
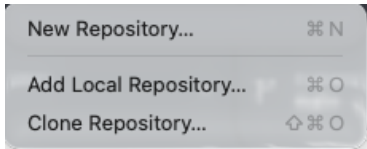
- Free desktop app for Windows and macOS
- Visual interface for common Git operations
- Easy branching and merging
- Automatic sync with GitHub
- Download from: <https://desktop.github.com/>

## IDE Integration:

- VS Code, JetBrains IDEs have built-in Git support

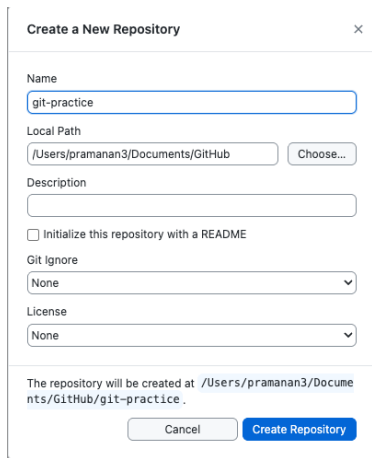
## Creating a Repository

# Creating a New Repository: Two Ways In



- **File menu:** File → New Repository...
- **Sidebar:** Click the + button next to "Current Repository"

# The New Repository Dialog

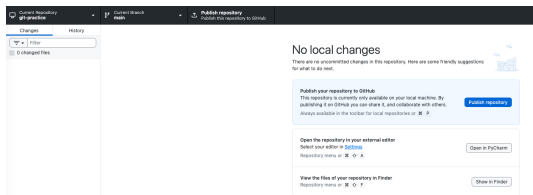


The screenshot shows the 'Create a New Repository' dialog box. It has a title bar with a close button. The form contains the following fields and options:

- Name:** A text input field containing 'git-practice'.
- Local Path:** A text input field containing '/Users/pramanan3/Documents/GitHub' and a 'Choose...' button.
- Description:** An empty text input field.
- Initialize this repository with a README:** An unchecked checkbox.
- Git Ignore:** A dropdown menu with 'None' selected.
- License:** A dropdown menu with 'None' selected.
- Summary:** A text area containing 'The repository will be created at /Users/pramanan3/Documents/GitHub/git-practice.'
- Buttons:** 'Cancel' and 'Create Repository'.

- **Name:** folder/repo name
- **Local Path:** where it lives on disk
- **Initialize with README:** leave **unchecked** (mirrors a bare `git init`)
- **Git Ignore / License:** optional, leave as None for now

# Result: An Empty Local Repository



- "No local changes" means nothing is staged, i.e. nothing uncommitted is pending.
- GitHub Desktop would have already run `git init` and auto-committed a `.gitattributes` file as an "Initial commit". (FYI: `git init` on CLI does neither.)
- Repository exists only on this machine (local repository)
- **Publish repository** button will push it to GitHub (Don't click, we will come back to it).



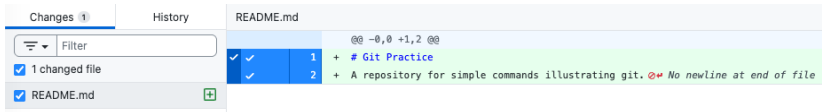
## Your First Commit

# Working Copy: Adding a File



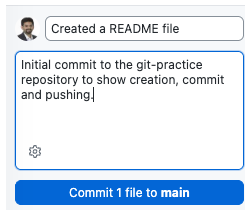
- Create a README.md in the repo folder using an editor
- This is a change in the **working copy** (i.e. your current folder). Git has not yet tracked it yet

# Staging Area: Selecting Changes



- Click on Changes and you should be able to see Checked file = staged.
- This is equivalent to `git add README.md`
- Green lines in the diff show what was added

# Writing the Commit



- Now we would need to make sure our changes are committed. So we will write a commit message.
- **Summary:** Typically a short, imperative statement, will be the commit message's first line
- **Description:** optional details on the *why*. (Ask yourself the question: Why was this change necessary?)
- **Click Commit 1 file to main** which is equivalent to CLI `git commit`

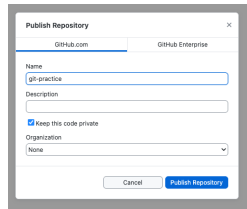
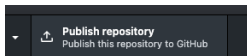
# Commit History

The screenshot shows the GitHub Desktop interface with the 'History' tab selected. The commit history list on the left shows two commits: 'Created a README file' (just now) and 'Initial commit' (18 minutes ago). The 'Created a README file' commit is selected, and its diff is shown on the right. The diff indicates that the README.md file was created, adding content about 'Git Practice'.

- Click on **History** tab to see what commits have been made (equivalent to `git log`)
- Click a commit to see its diff (i.e. what has changed, also equivalent to `git show`)
- This should show you two commits, README.md commit, plus the GUI's auto-generated "Initial commit"
- Both commits now live in the **local repository only!**

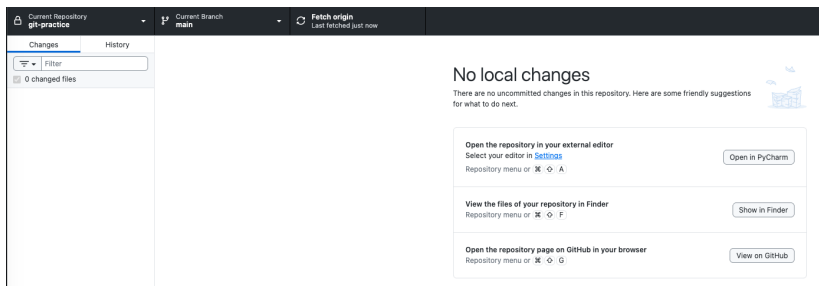
## Pushing to a Remote

# Publishing to GitHub



- **Publish repository** button appears when a repo only exists locally.
- You would need to push the repository to GitHub.com (i.e., the **remote**).
- Clicking "Publish" creates the **remote repository** on GitHub *and* pushes it.
- This is equivalent to `git remote add origin ...` followed by `git push -u origin main`

# Local and Remote in Sync

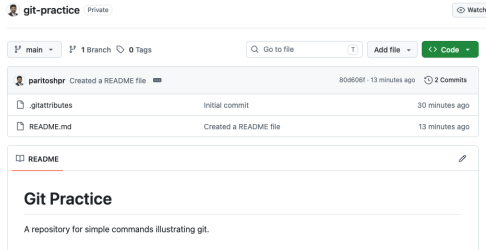


- Notice how **Publish repository** is now replaced by **Fetch origin**
- This means that your local repository and remote (GitHub) repository are now identical.



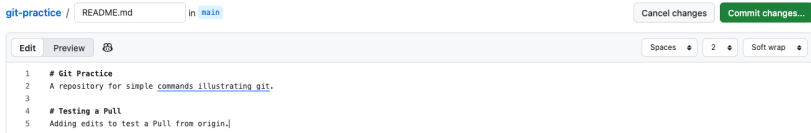
## Pulling from a Remote

# Simulating a Teammate's Change on GitHub.com



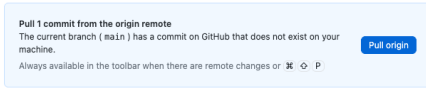
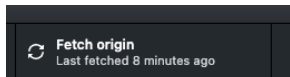
- Switch to GitHub.com's own web editor which is useful for quick edits without cloning.
- Editing README.md directly on GitHub.com and committing there simulates a change pushed by a teammate (or from another machine)
- Your local repository doesn't know about this change yet

# Editing on GitHub.com



- Add a new section and commit straight to `main` on GitHub.
- This is similar to our "Fork Commit and PR exercise in last class".
- This commit now exists only on the **remote repository**

# Fetch and Pull from GitHub Desktop



- Switch back to GitHub Desktop App.
- **Fetch origin**: checks the remote for new commits. (This is equivalent to `git fetch`)
- GitHub Desktop detects the remote is ahead and prompts you to **Pull origin** which is equivalent to `git pull`
- Fetch alone does not change your files; pull does

# Pull Complete

Current Repository: git-practice | Current Branch: main | Fetch origin: Last fetched 3 minutes ago

Changes | History

Edited README.md directly on GitHub

Adding edits to test Pull

Paritosh Ramanan · ce0242f

1 changed file | README.md

|   |   | @@ -1,2 +1,5 @@                                                                     |
|---|---|-------------------------------------------------------------------------------------|
| 1 | 1 | # Git Practice                                                                      |
| 2 | - | A repository for simple commands illustrating git. <i>No newline at end of file</i> |
|   | 2 | A repository for simple commands illustrating git.                                  |
|   | 3 | +                                                                                   |
|   | 4 | # Testing a Pull                                                                    |
|   | 5 | Adding edits to test a Pull from origin.                                            |

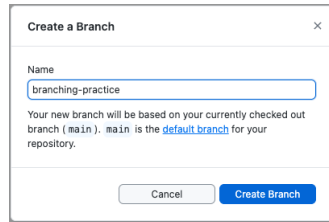
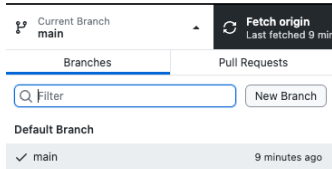
Created a README file  
Paritosh Ramanan · 18 minutes ago

Initial commit  
Paritosh Ramanan · 36 minutes ago

- The remote commit now appears in local **History**, with its diff visible
- Toolbar returns to **Fetch origin**. This means your local and remote are in sync again.

# Branching

# Creating a Branch (GitHub Desktop)



- **Current Branch** dropdown → **New Branch** (equivalent to `git branch + git checkout`)
- New branch is based on whatever branch you currently have checked out
- After you're done creating the new branch "branching-practice", you should automatically land on "branching practice" (right next to Publish branch).

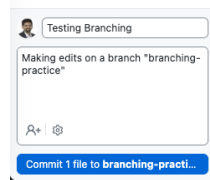
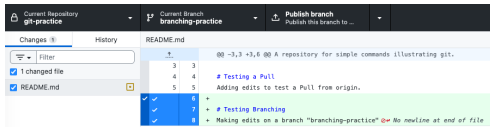
# Working on the Branch

```
1  # Git Practice
2  A repository for simple commands illustrating git.
3
4  # Testing a Pull
5  Adding edits to test a Pull from origin.
6
7  # Testing Branching
8  Making edits on a branch "branching-practice"
```

- Edit the raw file README.md while branching-practice is checked out. (You can use any IDE or text editor for this).
- Changes here do not affect main until merged



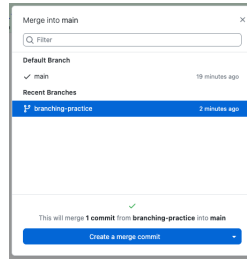
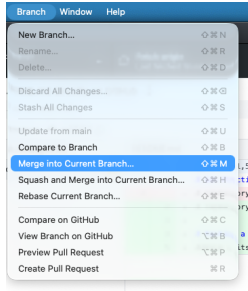
# Staging and Committing on a Branch



- Same stage/commit flow as before, but the toolbar confirms you're on branching-practice, not main

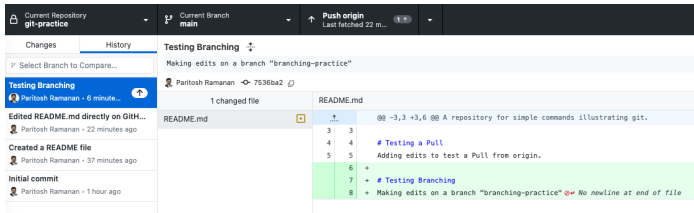
## Merging Into Main

# Merging: Two Ways In



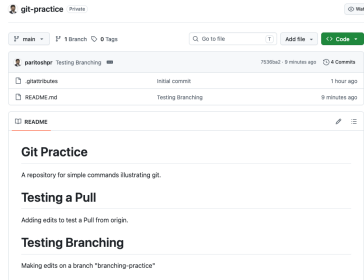
- Switch **Current Branch** back to `main` first from the top bar.
- **Branch** menu → **Merge into Current Branch...** – equivalent to `git merge branching-practice`
- Select the branch "branching-practice" to merge *into* `main` and confirm "Create a Merge Commit".

# Fast-Forward Merge



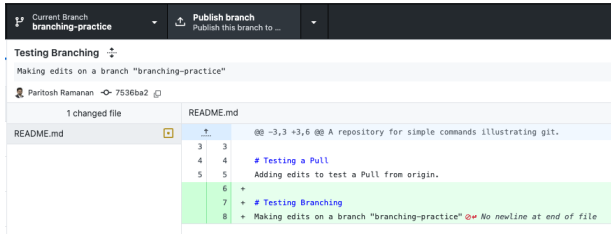
- main had no new commits since branching-practice was created, so Git just moves main forward.
- This is called a **fast-forward merge**, no separate merge commit
- Toolbar shows **Push origin [1↑]** which means this is ready to push to main. Click "Push Origin" at the top.

# Pushing the Merged main (Verify on Github.com)



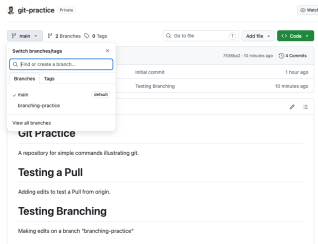
- After `git push`, GitHub's `main` shows the merged content – 4 commits total
- But notice: still **"1 Branch"** on GitHub.
- *The merge's content pushed, but the `branching-practice` branch itself never did*

# Publishing a Branch is a Separate Step (on GitHub Desktop)



- Switching back to branching-practice shows **Publish branch**, this branch was never pushed
- Merging and publishing a branch are independent actions

# The Branch, Now on the Remote

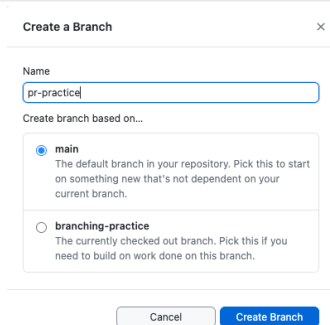


- GitHub now shows "**2 Branches**": main and branching-practice
- Closer to a typical team workflow: publish a branch, work on it, then merge – often via a **Pull Request** on GitHub rather than a local merge

## Pull Requests



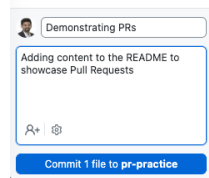
# A New Branch for the Pull Request (on GitHub Desktop)



- If you're still on branching-practice, you can again click on "Branch" → "New Branch".
- Same **New Branch** dialog, now with a choice: "**Create branch based on...**" main or branching-practice
- Basing on main keeps this change independent of the earlier one.

# Edit and Commit, as Before

```
M README.md x
1 # Git Practice
2 A repository for simple commands illustrating git.
3
4 # Testing a Pull
5 Adding edits to test a Pull from origin.
6
7 # Testing Branching
8 Making edits on a branch "branching-practice"
9
10 # Pull Request Practice
11 Make a pull request to let the admin merge onto main.
```



- As before, edit → stage → commit flow, this time on pr-practice with the files.

# Publish Before Requesting a Merge

## No local changes

There are no uncommitted changes in this repository. Here are some friendly suggestions for what to do next.



### Publish your branch

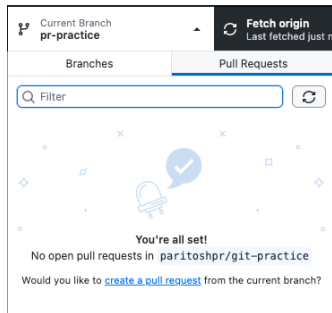
The current branch ( `pr-practice` ) hasn't been published to the remote yet. By publishing it to GitHub you can share it, open a pull request, and collaborate with others.

[Publish branch](#)

Always available in the toolbar or  **P**

- Unlike `branching-practice`, this branch ("`pr-practice`") is published *before* merging.
- A Pull Request needs the branch to exist on GitHub

# Starting a Pull Request from GitHub Desktop



- **Pull Requests** tab in the branch dropdown offers a shortcut. Click on "create a pull request" from the current branch.
- Opens your browser directly to GitHub's "Open a pull request" page

# Opening the Pull Request on GitHub (GitHub.com)

## Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also [compare across forks](#). [Learn more about diff comparisons here](#).

base: main ← compare: pr-practice ✓ Able to merge. These branches can be automatically merged.

 Add a title \*

Demonstrating PRs

Add a description

Write Preview H B I         

Adding content to the README to showcase Pull Requests

☐ Markdown is supported ☐ Paste, drop, or click to add files

Create pull request

Remember, contributions to this repository should follow our [GitHub Community Guidelines](#).

Reviewers  
No reviews

Assignees  
No one—[assign yourself](#)

Labels  
None yet

Projects  
None yet

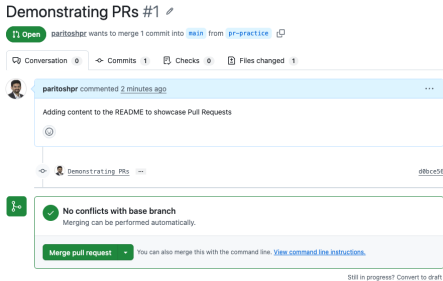
Milestone  
No milestone

Development  
Use [Closing keywords](#) in the description to automatically close issues

Helpful resources  
[GitHub Community Guidelines](#)

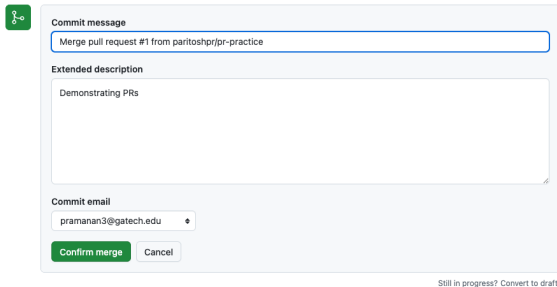
- **base: main** ← **compare: pr-practice**. Verify this at the very top, you should be able to see which branch merges into which
- Title and description, just like a commit message
- **Able to merge** confirms no conflicts

# Reviewing the Pull Request (GitHub.com)



- **Conversation, Commits, Checks, Files changed** tabs – this is where reviewers leave comments before merging
- In a team, a PR is usually reviewed and approved by someone *other* than its author before merging

# Merging the Pull Request (GitHub.com)



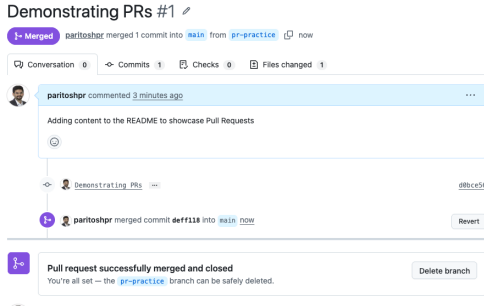
The screenshot shows a modal dialog for confirming a merge. It has a green GitHub logo icon in the top left. The dialog contains the following fields and buttons:

- Commit message:** A text input field containing "Merge pull request #1 from paritoshpr/pr-practice".
- Extended description:** A text area containing "Demonstrating PRs".
- Commit email:** A dropdown menu showing "pramanan3@gatech.edu".
- Buttons:** A green "Confirm merge" button and a grey "Cancel" button.

Below the dialog, there is a small text link: "Still in progress? [Convert to draft](#)".

- Clicking **Merge pull request** asks you to confirm a merge commit message
- This merge happens entirely on GitHub, your local repository doesn't know about it yet

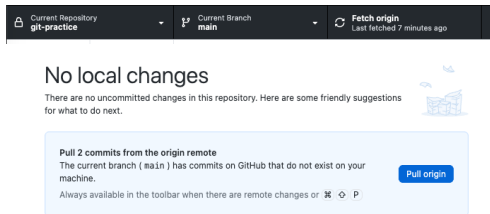
# Merged and Closed (GitHub.com)



- PR status changes to **Merged**
- GitHub offers **Delete branch** – once merged, pr-practice has served its purpose



# Bringing the Merge Back Locally (GitHub Desktop)



- Back on `main` locally, **Fetch origin** reveals **2 commits** waiting, the PR's commit and its merge commit
- **Pull origin** brings both down

# The Full Story, in History

| Changes                                                                  | History | Testing Branching ↕                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 Select Branch to Compare...                                            |         |                                                                                                                                                                                                                                                                         |
| Merge pull request #1 from paritosh...<br>Paritosh Raman · 3 minutes ago |         |                                                                                                                                                                                                                                                                         |
| Demonstrating PRs<br>Paritosh Raman · 8 minutes ago                      |         |                                                                                                                                                                                                                                                                         |
| Testing Branching<br>Paritosh Raman · 31 minutes ago                     |         |                                                                                                                                                                                                                                                                         |
| Edited README.md directly on GitHub...<br>Paritosh Raman · 1 hour ago    |         |                                                                                                                                                                                                                                                                         |
| Created a README file<br>Paritosh Raman · 1 hour ago                     |         |                                                                                                                                                                                                                                                                         |
| Initial commit<br>Paritosh Raman · 1 hour ago                            |         |                                                                                                                                                                                                                                                                         |
|                                                                          |         | Making edits on a branch "branching-practice"<br>Paritosh Raman · 7536ba2                                                                                                                                                                                               |
| 1 changed file                                                           |         | README.md                                                                                                                                                                                                                                                               |
|                                                                          |         | <pre> @@ -3,3 +3,6 @@ A repository for simple commands illustrating git. 3 3 4 4 # Testing a Pull 5 5 Adding edits to test a Pull from origin. 6 + 7 + # Testing Branching 8 + Making edits on a branch "branching-practice" No newline at end of file           </pre> |

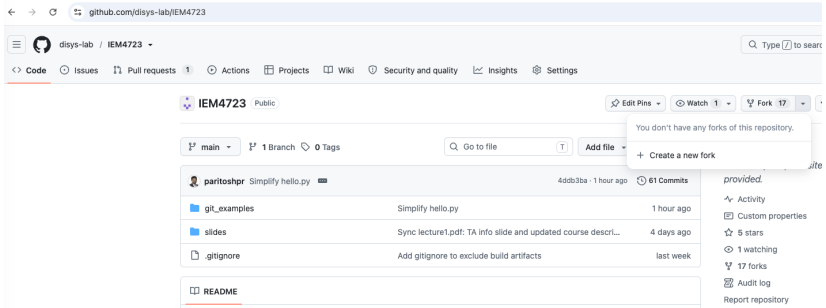
- Every step is visible: initial commit, README edits, the local branch merge, and now the **Pull Request** merge
- Local and remote history now match again

## Forking, Revisited

# What Is a Fork?

- A fork is your own copy of *someone else's* repository, under your account
- Unlike a branch, a fork is a completely separate repository – with its own remote
- You cannot fork your own repository into your own account
- This is how you'll get your own copy of the course repository

# Forking the Course Repository (GitHub.com)



- On dissys-lab/IEM4723 (the real course repo), click **Fork**
- "You don't have any forks of this repository", click **Create a new fork**

# The Create a New Fork Page (GitHub.com)

## Create a new fork

A fork is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project. [View existing forks.](#)

Required fields are marked with an asterisk (\*).

Owner \*

 paritoshpr

Repository name \*

ITEM4723

ITEM4723 is available.

By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.

Description

0 / 350 characters

☒ Copy the [main](#) branch only

Contribute back to [displays-lab/ITEM4723](#) by adding your own branch. [Learn more.](#)

 You are creating a fork in your personal account.

Create fork

- **Owner:** your personal account. FYI: GitHub won't let you fork into an account that already owns the repo
- **"Copy the main branch only"**. Fine for course purposes for now, keeps the fork lean

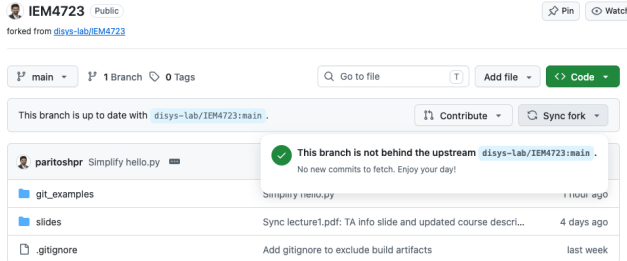
# Your Fork (GitHub.com)

The screenshot shows a GitHub repository page for a user named 'IEM4723' (Public). It indicates the repository was forked from 'disys-lab/IEM4723'. The interface shows the 'main' branch is selected, with 1 branch and 0 tags. A search bar and 'Go to file' button are present. A green 'Code' button is visible. A message states 'This branch is up to date with disys-lab/IEM4723:main'. Below this, there are 'Contribute' and 'Sync fork' buttons. A list of files is shown, including 'git\_examples', 'slides', and '.gitignore', each with a description and a commit hash/age.

| File         | Description                                                   | Commit               |
|--------------|---------------------------------------------------------------|----------------------|
| git_examples | Simplify hello.py                                             | 4ddb3ba · 1 hour ago |
| slides       | Sync lecture1.pdf: TA info slide and updated course descri... | 1 hour ago           |
| .gitignore   | Add gitignore to exclude build artifacts                      | 4 days ago           |

- **"forked from disys-lab/IEM4723"** label confirms the relationship
- **"This branch is up to date with disys-lab/IEM4723:main"** for now
- Clone *this* repository (your fork) in GitHub Desktop to work locally, exactly like git-practice

# Staying in Sync: Sync Fork (GitHub.com)

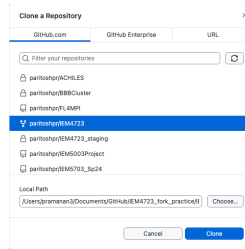
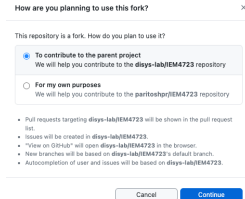


- As new material is pushed to disys-lab/IEM4723, your fork falls behind
- **Sync fork** pulls upstream changes into your fork on GitHub – equivalent to `git pull upstream main` + `git push`, done server-side
- Do this *always before* pulling to your local machine



**Putting It All Together:**  
**Fork → PR → Merge**

# Cloning Your Fork (GitHub Desktop)

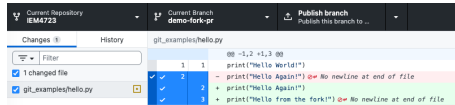
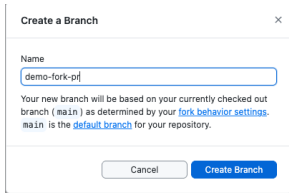


- The newly forked repository doesn't exist yet locally. Go ahead and click on "Current Repository → Add → Clone Repository"
- GitHub Desktop asks how you'll use the fork – **"To contribute to the parent project"** bases new branches on `disys-lab/IEM4723`'s default branch
- Clone `paritoshpr/IEM4723` from the **GitHub.com** tab, same as any repo

# Why Branch Before Opening a PR? (GitHub Desktop)

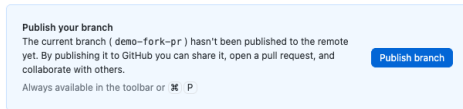
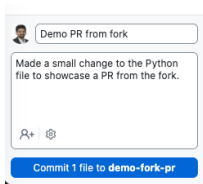
- Not strictly required. GitHub allows a PR from any branch, including `main`
- But committing directly to your fork's `main` makes it diverge from upstream, complicating future **Sync fork** operations
- A dedicated branch keeps `main` a clean mirror of upstream, and is easy to discard if the PR is never merged
- Standard convention is one branch per contribution

# Branch, Edit, Commit – On the Fork (GitHub Desktop)



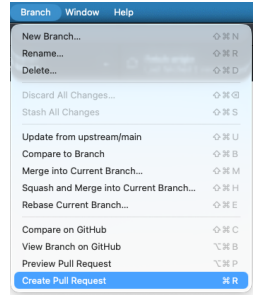
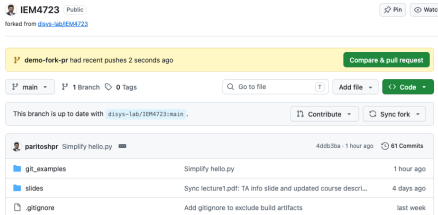
- Same flow as before. New branch demo-fork-pr, edit a file, stage, commit
- Notice the note: branch is based on main "as determined by your fork behavior settings". This was the choice we made when cloning

# Publish the Branch to Your Fork (GitHub Desktop)



- Commit, then **Publish branch** – pushes demo-fork-pr to *your fork*, paritoshpr/IEM4723, not to disys-lab/IEM4723

# Opening the PR: Two Ways In



- GitHub.com's **"Compare & pull request"** banner, or
- GitHub Desktop's **Branch → Create Pull Request** – both land on the same page

# A Cross-Repository Pull Request

## Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also [compare across forks](#). [Learn more about diff comparisons here](#).

base repository: disys-lab/IEM4723 | base: main | head repository: paritoshpr/IEM4723 | compare: demo-fork-pr

✓ Able to merge. These branches can be automatically merged.

**Add a title \***

**Add a description**  

Write | Preview

Made a small change to the Python file to showcase a PR from the fork.

Markdown is supported | Paste, drop, or click to add files

☒ Allow edits by maintainers | **Create pull request**

Remember, contributions to this repository should follow our [GitHub Community Guidelines](#).

**Reviewers**  
No reviewers

**Assignees**  
No one—assign yourself

**Labels**  
None yet

**Projects**  
None yet

**Milestone**  
No milestone

**Development**  
Use [closing keywords](#) in the description to automatically close issues

**Helpful resources**  
[GitHub Community Guidelines](#)

- This is the key difference from our earlier PR: **base repository: disys-lab/IEM4723** ← **head repository: paritoshpr/IEM4723**
- Two separate repositories, not just two branches in one repo

# Merging a Fork's Pull Request



No conflicts with base branch

Merging can be performed automatically.

Merge pull request

You can also merge this with the command line. [View command line instructions.](#)

Still in progress? [Convert to draft](#)



Commit message

Merge pull request #15 from paritoshpr/demo-fork-pr

Extended description

Demo PR from fork

Commit email

pramanan3@gatech.edu

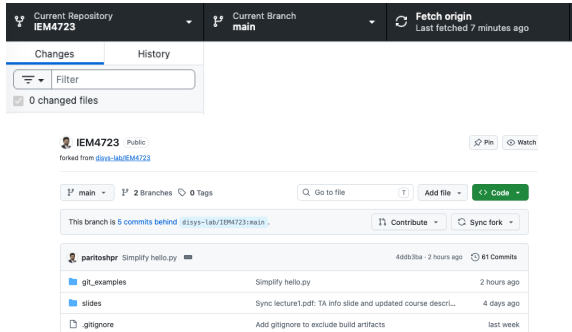
Confirm merge

Cancel

- Same **Merge pull request** → **Confirm merge** flow as before
- This merge lands in `disys-lab/IEM4723` – the *upstream* repository

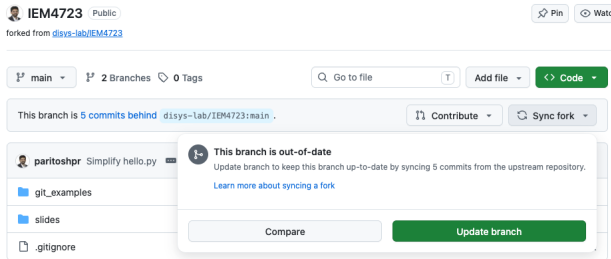


# The Gotcha: Your Fork Doesn't Auto-Update



- Merging your PR updates `disys-lab/IEM4723`, **not** `paritoshpr/IEM4723`
- Your fork is now "**5 commits behind**" – your merge, plus others' merges (like a classmate's PR)

# Sync Fork, Then Pull Locally



- **Sync fork** → **Update branch** brings your fork's main up to date with upstream
- Then **Fetch origin** / **Pull origin** in GitHub Desktop brings it to your machine

## Full Circle

The screenshot shows the GitHub Desktop interface. At the top, it displays the current repository (IEMA723), current branch (main), and a 'Fetch origin' button. The sidebar on the left shows a list of changes, including 'Merge pull request #14 from ttvzi...', 'Merge pull request #15 from paritoshpr/demo-fork-pr', 'Demo PR from fork', and 'Simplify hello.py'. The main area shows the diff for the selected pull request, highlighting changes in the 'git\_examples/hello.py' file. The diff shows three lines of code being added or modified, with line numbers 1, 2, and 3. The bottom section, titled 'Commits', shows a list of commits, including 'Merge pull request disys-lab#14 from ttvzicotix/main' and 'Merge pull request disys-lab#15 from paritoshpr/demo-fork-pr'.

- Your merged PR appears in local history, alongside a classmate's merged PR (#14) – everyone's contributions, in one shared history
- Fork → branch → commit → push → PR → merge → sync → pull – the complete collaboration workflow